

Windows Network Troubleshooting & Diagnostics Lab

Portfolio Project

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Focus: IT Support | Help Desk | Windows Networking

Environment: Windows 11

Type: Hands-on network diagnostics and troubleshooting practice

1. Project Overview

I created this project to build practical IT Support and Windows networking troubleshooting skills. My goal was not to memorize commands. I wanted to understand what to test, why I am testing it, how to read the result, and what to do next.

The project covers five common scenarios a Help Desk technician may face:

- DHCP and APIPA addressing
- DNS resolution problems
- Default gateway connectivity
- Connected to Ethernet but no internet
- IP address conflicts

Important note: I ran the diagnostics on a working organizational workstation, not an isolated virtual machine. I did not change or break the organization's network. I recorded healthy baselines, ran safe diagnostic tests, and documented how I would recognize, isolate and troubleshoot each type of fault.

2. Tools and Commands

Command	Purpose
ipconfig / ipconfig /all	Review IP configuration, DHCP, gateway and DNS
ping	Test connectivity to the gateway, an external IP and a hostname
nslookup	Test DNS resolution, including a direct query to a DNS server
tracert	Observe the network path to an external destination
arp -a	Inspect the local IP-to-MAC address cache

I also studied `ipconfig /release` and `ipconfig /renew` but did not run them on the organizational workstation, because the DHCP configuration was healthy and disrupting it was unnecessary.

3. Troubleshooting Methodology

I worked outward from the workstation, one layer at a time:

1. Check physical / network connection
2. Check IP and DHCP configuration
3. Identify the default gateway
4. Ping the gateway (local connectivity)
5. Ping an external IP (external connectivity)
6. Test DNS resolution
7. Trace the network path
8. Test browser / application access
9. Inspect ARP information if required
10. Interpret results, choose next action, document findings

4. Diagnostic Scenarios

Scenario 01: DHCP and APIPA

User report: "My computer is connected to the network, but I cannot access network resources or the internet."

Test: `ipconfig /all`

Observations (addresses sanitized):

```
DHCP Enabled : Yes
IPv4 Address : 192.168.x.x
Subnet Mask : 255.255.254.0
Default GW : 192.168.x.x
DHCP Server : 192.168.x.x
DNS Servers : Internal DNS servers
```

Analysis: The workstation received a valid private IPv4 address through DHCP, with subnet mask, gateway, DHCP server and DNS all present.

APIPA indicator: An address in the 169.254.x.x range on a DHCP client means the workstation did not get its configuration. I would then check the Ethernet or Wi-Fi connection, adapter status, DHCP availability and configuration, and the network infrastructure. The 169.254.x.x address is a symptom. The real task is finding out why DHCP was not obtained.

Recovery knowledge: In an authorized environment, `ipconfig /release` followed by `ipconfig /renew` refreshes DHCP information. I did not run it here.

Gateway test: `ping 192.168.x.x` returned 4 sent, 4 received, 0% loss.

Result: Healthy DHCP configuration and successful gateway communication. No active DHCP/APIPA fault observed.

Scenario 02: DNS Resolution

User report: "I'm connected, but websites are not loading."

I separated IP connectivity from name resolution instead of assuming the internet was down.

Test	Command	Result
External IP connectivity	<code>ping 8.8.8.8</code>	4/4 replies, avg 13 ms - PASS
DNS resolution	<code>nslookup google.com</code>	IPv4 and IPv6 returned - PASS

Test	Command	Result
Direct DNS server query	nslookup google.com <DNS-server>	Internal DNS returned records - PASS
Hostname connectivity	ping google.com	4/4 replies, avg 12 ms - PASS

Analysis: If ping to an external IP works but hostname lookups keep failing, DNS becomes the area to investigate.

Result: No active DNS fault observed.

Scenario 03: Default Gateway

User report: "I can't access network resources or the internet."

I used `ipconfig` to identify the IPv4 address, subnet mask and default gateway, then ran `ping <default-gateway>`: 4 sent, 4 received, 0% loss.

If the gateway ping fails, I would investigate closer to the workstation first:

- Ethernet or Wi-Fi connection, and the correct wireless network
- Network adapter
- IPv4, subnet and default gateway configuration

I then ran `tracert 8.8.8.8`. The trace reached `dns.google [8.8.8.8]` in 12 hops. One intermediate hop showed "Request timed out", but later hops responded and the trace completed.

Learning point: A single timeout in `tracert` does not mean the whole connection has failed. The full result must be interpreted.

Result: Valid gateway, successful gateway communication and a working path to the external destination. No active gateway fault observed.

Scenario 04: Connected to Ethernet but No Internet

User report: "My Ethernet says connected, but I can't access the internet."

I combined the earlier tests into one structured process:

Step	Test	Status
1	<code>ipconfig /all</code> : IP, mask, gateway, DHCP, DNS present	PASS
2	<code>ping default gateway</code> (0% loss)	PASS
3	<code>ping 8.8.8.8</code> (0% loss)	PASS
4	<code>nslookup google.com</code>	PASS
5	<code>ping google.com</code> (0% loss)	PASS
6	<code>tracert 8.8.8.8</code> (destination reached)	PASS
7	Browser test: <code>https://www.microsoft.com</code> loaded	PASS

Evidence: [Insert screenshot here]

Figure 1: Browser connectivity verification after validating IP configuration, gateway connectivity, external connectivity, DNS resolution and routing.

Result: No active "connected but no internet" fault observed. The key outcome was learning to break a broad complaint into smaller diagnostic questions and find which layer works and where a failure would begin.

Scenario 05: IP Address Conflict

User report: "My connection keeps dropping and Windows reports an IP address conflict."

A conflict needs careful investigation. Randomly assigning a new IP may hide the problem instead of fixing it. My workstation uses DHCP (DHCP Enabled: Yes), so no static address was being tested.

I ran `arp -a` to view the local ARP cache. Public output is sanitized:

Interface: 192.168.x.x		
Internet Address	Physical Address	Type
192.168.x.1	[REDACTED]	dynamic
192.168.x.x	[REDACTED]	dynamic

Learning point: ARP entries alone do not prove an IP conflict. Results must be read together with other evidence.

How I would investigate a suspected conflict (authorized environment):

- Run `ipconfig /all`: check IPv4 address, DHCP status, DHCP server, gateway and adapter
- Confirm whether the address is automatic (DHCP) or manually configured
- Inspect ARP information for the IP-to-MAC mappings
- Use authorized network-management resources to see whether another device holds the address
- Fix the identified root cause instead of guessing
- Verify IP configuration, gateway connectivity, DNS and access to the affected resource

Result: No active IP conflict identified. I gained practice reading an ARP table and understanding how IP address, ARP, MAC address and local device relate.

5. Overall Results

Scenario	Status
01 DHCP / APIPA	Diagnostic completed
02 DNS Resolution	Diagnostic completed
03 Default Gateway	Diagnostic completed
04 Connected but No Internet	Diagnostic completed
05 IP Address Conflict	Diagnostic completed

"Completed" means the diagnostic exercise was done. It does not mean a fault was created or repaired. The environment was healthy, and I did not claim to have fixed failures that did not occur.

6. Key Lessons Learned

1. Do not guess. Good troubleshooting needs evidence. For every test I ask what it tells me if it passes, what it tells me if it fails, and what to test next.

2. Start close to the workstation. If the PC cannot reach its gateway, testing external services is premature. Work outward: workstation, IP configuration, local network, gateway, external network,

DNS, application.

3. DNS and internet connectivity are not the same. Testing an external IP separately from a hostname shows which type of failure is present.

4. A symptom is not always the root cause. A 169.254.x.x address points the investigation. It does not explain why DHCP failed.

5. Read results in context. One timeout in tracert did not mean failure because the trace still completed.

6. Protect the production environment. On an organizational workstation I chose not to disable DHCP, assign conflicting IPs, set incorrect DNS servers, delete ARP entries, disrupt adapters, or introduce any fault. Good IT Support includes understanding risk, authorization and when a change is unnecessary.

7. Skills Demonstrated

Area	Skills
IT Support	L1 troubleshooting, fault isolation, root-cause thinking, technical documentation, user-focused troubleshooting
Windows	Command Prompt, network configuration, network adapters, TCP/IP diagnostics
Networking	IPv4, subnet masks, default gateways, DHCP, APIPA, DNS, ARP, MAC addressing, ICMP testing, routing
Tools	ipconfig, ping, nslookup, tracert, arp
Professional	Attention to detail, documentation, logical thinking, responsible change management, willingness to learn

8. Reflection and Next Steps

This project moved me from knowing that commands like ping and ipconfig exist to understanding when to use each one and how the result shapes the next decision. Good troubleshooting is knowing what to test, why, how to read the result, and what to do next.

I received feedback that a healthy baseline should not be described as resolving a real incident. I restructured the project to document it accurately as diagnostic scenarios.

My next step is hands-on labs in an isolated environment covering Windows administration, Active Directory, service desk operations and networking, where I can create real faults and document the fixes.

Documentation note: Before publishing, internal domain names, hostnames, exact addresses, DNS server details and MAC addresses are removed or anonymized. The public version focuses on methodology and skills, not the organization's infrastructure.

GitHub repository: [Windows-Network-Troubleshooting-and-Diagnostics-Lab](#)